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Linking Hardness Measurements to Brittleness Index for Unconventional Reservoir Characterization: Insight from the Horn River Basin, Northeast British Columbia, Canada

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Assessing the prospectivity of unconventional reservoirs presents several challenges (Wang et al., 2016). These include: (1) complex geological characteristics of the target formations, (2) selection of appropriate and relevant evaluation parameters and methods, and (3) identification of intervals with high potential for resource extraction. A key step in this process is determining the most prospective zones—commonly called sweet spots—within an unconventional play. The assessment substantially reflects two parameters: total organic carbon (TOC) and brittleness. TOC represents the abundance of organic matter within sedimentary source rocks that has the potential to generate hydrocarbons under elevated temperatures, while brittleness reflects the formation's capacity to undergo effective natural or hydraulic fracturing—an essential property for enhancing permeability in unconventional reservoirs, which are typically characterized by ultralow permeability (Iyare et al., 2021; Ore & Gao, 2021). An ideal sweet spot combines high TOC content with sufficient brittleness, improving fracability (Yuan et al., 2017).

A common challenge in estimating continuous TOC and brittleness profiles lies in the subjectivity involved in interpolating and extrapolating sparse laboratory measurements (from core samples, cuttings, etc.) to define consistent trends or reference values across the entire well. Traditional laboratory methods

such as Rock-Eval pyrolysis (e.g., [Barker, 1974](#)) and LECO induction furnace analysis (e.g., [Law, 1999](#)) are widely used for TOC estimation. However, these techniques rely on laboratory measurements from core or cutting samples. While cuttings are more widely available than cores, both are still limited to discrete depth intervals and lack the continuous resolution needed for full-well characterization. To overcome this limitation, empirical approaches based on well logs (e.g., [Passey et al., 1990](#); [Schmoker, 1979](#)) have been developed to estimate TOC continuously along the wellbore. More recently, artificial intelligence (AI) and machine learning (ML) techniques (e.g., [Davy et al., 2024a,b](#)) have also been applied to generate continuous TOC profiles with improved accuracy. The aforementioned study ([Davy et al., 2024a](#)) incorporated a petrophysical constraint called δ , derived from the difference between neutron porosity and bulk density, which effectively distinguished clay-rich (non-prospective) from clay-lean (prospective) intervals. A similar parameter, Δ , inspired by [Hall et al. \(2016\)](#) and based on the difference between neutron porosity and density porosity, exhibited comparable behavior

Brittleness estimation encounters difficulties similar to TOC, particularly due to limited core data and the subjectivity in establishing reference trends that interpolate or extrapolate between sparse measurement points. The Brittleness Index (BI), often used for this purpose, is difficult to generalize from sparse core samples alone, prompting empirical methods. [Mews et al. \(2019\)](#) categorize brittleness estimation into three main approaches: mineralogical, log-based, and elastic-based. The mineralogical-based brittleness index (MBI) links higher quartz content to greater brittleness and higher clay content to reduced brittleness, while the effects of calcite and dolomite vary across models. The log-based brittleness index (LBI) typically uses conventional well logs (e.g., neutron porosity, gamma ray), while the elastic-based brittleness index (EBI) relies on elastic parameters such as Poisson's ratio and Young's modulus, which may be derived from either well logs (e.g., dipole sonic and density) or laboratory measurements. While this difference may overlap in practice, particularly since elastic properties may often be derived from logs, this classification calls attention to different data types and modelling assumptions. [Mews et al. \(2019\)](#) state that a universal brittleness model is unrealistic; therefore, they favor the MBI as the one that represents the intrinsic property of the rock's material.

Introduction

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Hardness is another important parameter closely related to mineral composition and mechanical properties, and it may correlate with brittleness. Lawn & Marshall (1979) defined brittleness as the ratio of hardness (resistance to deformation) to toughness (resistance to fracture), showing a linear relationship between hardness and brittleness. Altindag & Guney (2010) found a strong correlation between hardness and brittleness index (BI), where BI was derived from uniaxial compressive and tensile strength. Dong et al. (2017) examined the relationship between hardness and brittleness on the same data as this study. They found that clay content is the main controlling factor. Quartz content also plays a role, but its effect depends on the type: authigenic quartz increases brittleness, while detrital quartz has little or no impact. More recently, Kim et al. (2022) studied rocks from the Liard Basin, adjacent to the Horn River Basin, examined in the current study. They reported that hardness correlates well with brittleness, making it a useful proxy.

Although TOC, brittleness, and hardness are essential, conventional techniques to obtain these parameters from core samples are time-consuming and limited. In the case of brittleness alone, the issue is even more severe since no single technique works optimally for all datasets. This represents a serious gap in accurately and effectively predicting these properties in other wells, depths, or paleoenvironments. In addition, the relationship between hardness and brittleness is complex and is a function of mechanical and lithological variations—hardness depends on rock mineralogy and geometry, while brittleness is a function only of volumetric attributes, so generalization is complex. We use cross-plot analysis to examine their relationship and gain some insight.

This research aims to address three main objectives: (1) compare hardness with different brittleness indices—LBI, MBI, and EBI; (2) examine the relationship between hardness and brittleness in the context of unconventional reservoir evaluation; and (3) identify key factors and well log responses that influence the correlation between TOC, hardness, and brittleness.

In addition, this study introduces a new validation method by using hardness testing to assess brittleness indices. This provides a practical and physically meaningful approach to brittleness estimation. With MBI as the most reliable indicator, the study offers improved understanding and practical insights for characterizing unconventional reservoirs.

The study confirms that the MBI is the best BI for our data, with the highest correlation coefficient with hardness, and the worst is the EBI. The MBI-hardness crossplot could be a prospective tool for characterizing unconventional prospects, identifying trends in organic matter and clay content. Quartz and clay content significantly influence the MBI-hardness relationship, while carbonate has minimal impact.

Materials and methods

This study uses data from five wells in the Horn River Basin, Northeast British Columbia, Canada, listed from south to north: (a) ArcSnake, (b) Komie, (c) McAdam, (d) Nexen Gote, and (e) Maxhamish (Fig. 1). Each well includes three key stratigraphic units, in order from oldest to youngest: the carbonate-rich Evie Member, the clay-rich Otter Park Member, and the silica-rich Muskwa Formation. The dataset comprises 6,366 points, with 612 TOC and 3,492 hardness measurements. The workflow of Davy et al. (2024a) was used to get the whole 6366 data points for the TOC and hardness (Figs. 2a and 2b, respectively), which will be used in this study. The method incorporates a petrophysical feature, delta (δ), defined as the difference between neutron porosity (ϕ_N) and bulk density (ρ_b), shown to be effective in predicting TOC and hardness. A similar parameter from Hall (2016), denoted as cap delta (Δ), is also used and defined as the difference between ϕ_N and density porosity (ϕ_D). Table 1 summarizes both δ and Δ , along with other well log data.

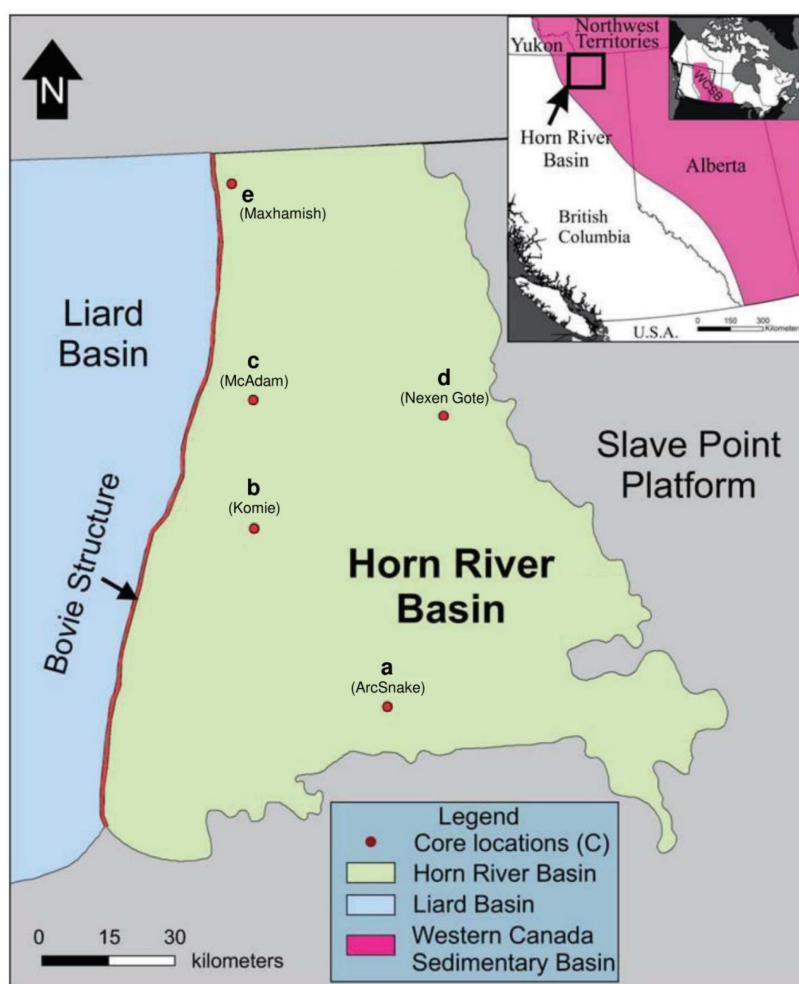
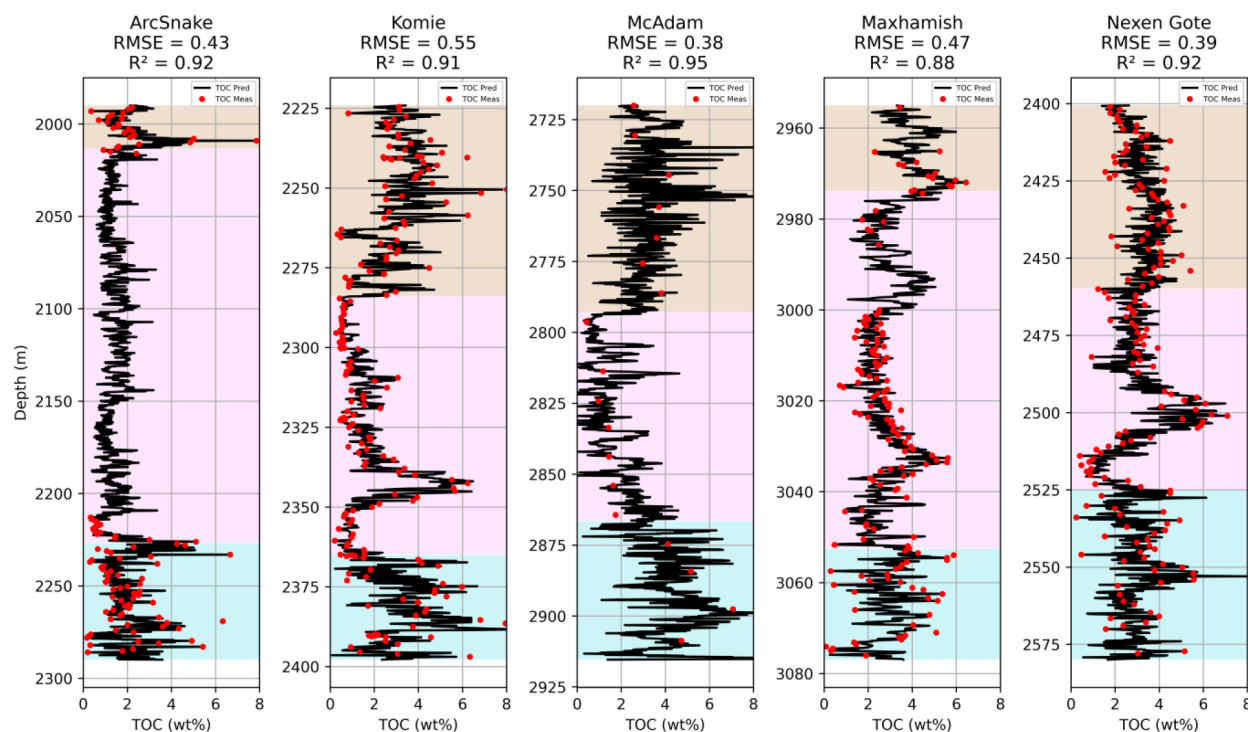


Figure 1—The five wells of the Horn River Basin, British Columbia, Canada, used in the study are shown from south to north; the wells are (a) ArcSnake, (b) Komie, (c) McAdam, (d) Nexen Gote, and (e) Maxhamish. Modified after Ayraanci et al. (2018a,b) and Davy et al. (2024a,b).

(a) The TOC references based on the modified workflow of Davy et al. (2024a)



(b) The hardness references based on the modified workflow of Davy et al. (2024a)

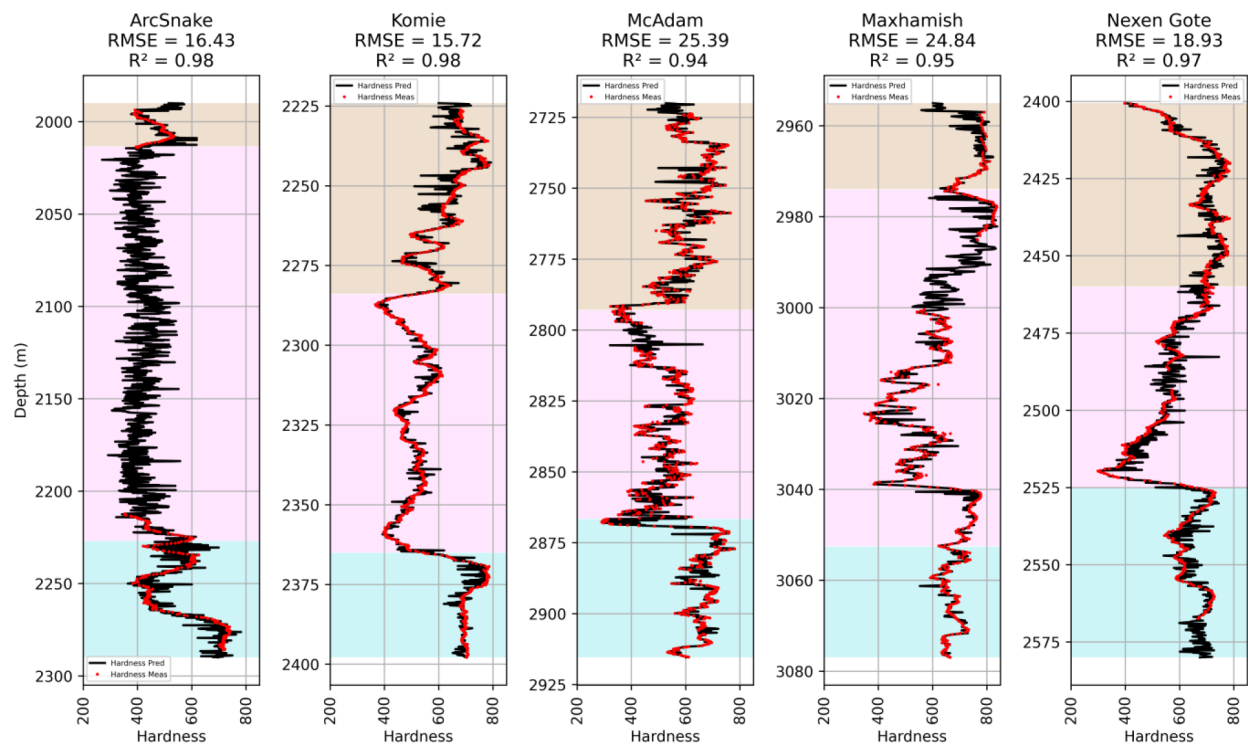


Figure 2—The references for (a) TOC and (b) hardness estimations based on the workflow of Davy et al. (2024a) for each of the wells. The RMSE (root-mean-square error) and R^2 (coefficient of determination) values are included. Red dots represent measured (actual) values, and black curves represent predicted values. Background colors indicate stratigraphic units of prospective intervals in the basin: brown for the Muskwa formation, magenta for the Otter Park member, and cyan for the Evie member, from the most recent to the oldest.

Table 1—Summary of 14 well logs and 2 petrophysical features statistics for five wells, including gamma ray (GR), thorium count (Th), potassium count (K), uranium count (U), bulk density (ρ_b), neutron porosity (ϕ_N), density porosity (ϕ_D), resistivity logs (R20, R30, and R90 for shallow, medium, and deep respectively), log10-transformed resistivity logs (log10_R20, log10_R30, and log10_R90), photoelectric effect (PE), ϕ_N - ρ_b crossover difference (δ), and ϕ_N - ϕ_D crossover difference (Δ).

Well Logs / Petrophysical Features	GR [gAPI]	Th [ppm]	K [%]	U [ppm]	RhoB [g/cm3]
Mean	121.574	7.893	1.833	8.295	2.620
STD	44.413	3.179	0.843	6.981	0.115
Min	9.277	0.373	0.055	0.002	2.342
Q1	98.129	5.450	1.112	2.292	2.524
Q2	110.798	7.704	1.696	6.957	2.605
Q3	141.808	10.699	2.606	12.563	2.726
Max	407.888	23.702	4.144	48.029	2.873
Well Logs / Petrophysical Features	NPHI [Frac.]	DPHI [Frac.]	R20 [ohm.m]	R30 [ohm.m]	R90 [ohm.m]
Mean	0.127	0.058	96.199	72.494	64.558
STD	0.051	0.073	279.161	161.890	147.191
Min	-0.011	-0.095	1.474	1.471	1.702
Q1	0.088	-0.009	11.236	11.052	10.408
Q2	0.129	0.064	19.605	19.143	17.806
Q3	0.166	0.116	51.527	51.158	45.669
Max	0.429	0.350	1950.000	1859.945	1950.000
Well Logs / Petrophysical Features	log10_R20 [log10(ohm.m)]	log10_R30 [log10(ohm.m)]	log10_R90 [log10(ohm.m)]	PE [b/e]	δ [Frac.]
Mean	1.444	1.426	1.388	4.271	0.109
STD	0.564	0.537	0.527	0.602	0.102
Min	0.169	0.167	0.231	2.592	-0.077
Q1	1.051	1.043	1.017	3.822	0.017
Q2	1.292	1.282	1.251	4.269	0.093
Q3	1.712	1.709	1.660	4.613	0.210
Max	3.290	3.270	3.290	6.618	0.396
Well Logs / Petrophysical Features	Δ [Frac.]	DTC [μsec/m]	DTS [μsec/m]	V_p [km/s]	V_s [km/s]
Mean	0.069	241.851	399.395	4.164	2.531
STD	0.105	19.891	43.023	0.355	0.257
Min	-0.213	172.242	318.535	3.261	1.831
Q1	-0.025	231.126	367.155	3.955	2.360
Q2	0.052	241.365	391.046	4.143	2.557
Q3	0.174	252.857	423.779	4.327	2.724
Max	0.362	306.618	546.245	5.806	3.139

Brittleness indices (BIs) are calculated based on data availability. The elastic brittleness index (EBI), which uses P-wave velocity (V_p) and S-wave velocity (V_s) derived from DTC and DTS (shear sonic) logs, is computed only for the Komie, McAdam, and Maxhamish wells—these are the only wells with both logs. The mineralogical brittleness index (MBI), based on normalized mineral weight data, is calculated for the ArcSnake and McAdam wells, where such data are available. The log-based brittleness index (LBI) is derived from ϕ_N logs and is available for all five wells; however, the version of LBI requiring DTC data is limited to the same three wells as EBI. As a result, BI values are restricted to wells with the necessary input data. If EBI, MBI, or DTC-based LBI proves most relevant to hardness, we will apply AI-based curve fitting using Davy et al. (2024a, b) to estimate BI values across all wells. Table 2 summarizes the equations for LBI, EBI, and MBI from Mews et al. (2019).

The work is divided into three stages: (I) Data preparation; (II) Feature selection and BI reference estimation; and (III) Cross-plot analysis. The details for each stage are discussed in the following subsections.

Table 2—Summary of brittleness indices (BI), including mineralogical-based (MBI), log-based (LBI), and elastic-based (EBI), as derived from Mews et al. (2019). Notes are also taken from this reference with additional sources. Each BI is calculated, and their relevance (coefficient correlation or R -value) to hardness is shown in Fig. 4.

Formula	Reference	Notes	Denoted in this study as
$\frac{Q}{Q + Carb + Cly}$	Jarvie et al. (2007)	Formation: Barnett. This equation yields accurate results for quartz- and clay-rich formations with low carbonate content, but it underestimates brittleness when the carbonate content is significant. Q , $Carb$, and Cly are the weight percentages of quartz, carbonate, and clay, respectively.	MBI 1
$\frac{Q + F + M + Carb}{tot}$	Jin et al. (2014a)	Formation: Barnett. Besides carbonate and quartz, feldspar and mica are also considered brittle minerals. All minerals, both brittle and non-brittle, should be included in the denominator. This equation is applicable to various lithologies with different predominant minerals. Q , F , M , $Carb$, and Cly are the weight percentages of quartz, feldspar, mica, carbonate, and clay, respectively.	MBI 2
$\frac{Q + F + M + Carb}{tot + \phi_{tot}}$	Jin et al. (2014a), modified by Mews et al. (2019)	Formation: Barnett. Added porosity in the denominator. Mews et al. (2019) used total porosity (ϕ_{tot}), but in our study, neutron porosity (ϕ_N) yielded a better correlation (R -value) with hardness compared to density and total porosity (ϕ_D and ϕ_{tot} , respectively). Q , F , M , and $Carb$ are the weight percentages of quartz, feldspar, mica, and carbonate, respectively. tot means all the weight percentages, including clay.	MBI 3
$-0.012 \times DTC + 1.4921$	Jin et al. (2014b)	Formation: Woodford. Estimate brittleness with limited data but handle these empirical equations carefully when applied to other formations. ϕ_N represents neutron porosity, while DTC denotes compressional sonic log values, measured in $\mu s/ft$. Similar note for the rest of the LBIs	LBI 1
$-1.5314 \times \phi_N + 0.8575$			LBI 2
$-0.01104 \times DTC + 1.4941$		Formation: Barnett.	LBI 3

$-1.4956 \times \phi_N + 0.9763$			LB1 4
$-0.0116 \times DTC + 1.6231$		Formation: Eagle Ford.	LB1 5
$-2.3115 \times \phi_N + 1.0104$			LB1 6
$-0.0142 \times DTC + 1.7439$		Formation: Global Correlation.	LB1 7
$-1.8748 \times \phi_N + 0.9679$			LB1 8
$\frac{E_{stat,norm} + v_{stat,norm}}{2}$	Rickman et al. (2008)	Formation: Barnett. - Dynamic shear and bulk moduli (G_{dyn} and K_{dyn}) are obtained from P-wave velocity (V_p), S-wave velocity (V_s), and bulk density (ρ_b). V_p and V_s are obtained from compressional and shear sonic logs (DTC and DTS), respectively. $G_{dyn} = \rho_b \times V_s^2$ $K_{dyn} = \rho_b (V_p^2 - \frac{4}{3} V_s^2)$ - Dynamic Young's modulus and Poisson's ratio (E_{dyn} and ν_{dyn}) are calculated from G_{dyn} and K_{dyn} $E_{dyn} = \frac{9 K_{dyn} \times G_{dyn}}{3 K_{dyn} + G_{dyn}}$ $\nu_{dyn} = \frac{3 K_{dyn} - 2 G_{dyn}}{6 K_{dyn} + 2 G_{dyn}}$ - Convert E_{dyn} to E_{stat} (static Young's modulus) using equation from Mullen et al. (2007), and ν_{dyn} to ν_{stat} (static Poisson's ratio) using equation from Zhang and Li (2020): $E_{stat} = \left(\frac{E_{dyn}}{3.3674} \right)^{2.042}$ $\nu_{stat} = 1.243 \nu_{dyn} - 0.0911$ - Normalized both E_{stat} and ν_{stat} of each datapoints, based on the maximum and minimum values	EB1 1

		$E_{stat,norm} = \frac{E_{stat} - E_{stat,min}}{E_{stat,max} - E_{stat,min}}$ $v_{stat,norm} = \frac{v_{stat} - v_{stat,min}}{v_{stat,max} - v_{stat,min}}$	
$E_{dyn} \times \rho_b$	Sharma & Chopra (2012)	Formation: Western Canadian Basin. Despite it is considered as an "index" in Mews et al. (2019), however, the value could exceed 1. Therefore, this "index" is more acceptable to be considered as an "indicator". Similar note as for EBI 3 and EBI 4	EBI 2
$\frac{E_{dyn} \times \rho_b}{v_{dyn}}$	Sun et al. (2013)	Formation: Liahoe, China.	EBI 3
$\frac{E_{dyn}}{\lambda}$	Chen et al. (2014)	Formation: unspecified. λ is Lamé's first parameter, obtained as shown below. $\lambda = \rho_b \times (V_p^2 - 2V_s^2)$	EBI 4
$\frac{EBI_{Rickman et al.(2008),norm} + KIC_{norm}}{2}$	Jin et al. (2014a)	Formation: Barnett. Utilizing fracture toughness (KIC): $KIC = 0.313 + 0.027 \times E_{stat}$ Then normalized both KIC and EBI obtained from Rickman et al. (2008), similar way as obtaining $E_{stat,norm}$ and $v_{stat,norm}$.	EBI 5
$\frac{EBI_{Rickman et al.(2008),norm} + GC_{norm}}{2}$	Jin et al. (2014a)	Formation: Barnett. Utilizing the strain energy release rate (GC): $GC = (1 - v_{dyn}^2) \times \left(\frac{KIC^2}{E_{dyn}} \right) \times 10^3$	EBI 6
		Then normalized both GC and EBI obtained from Rickman et al. (2008), similar way as obtaining $E_{stat,norm}$ and $v_{stat,norm}$.	
$\frac{EBI_{Rickman et al.(2008),norm} + E_{dyn,norm}}{2}$	Jin et al. (2014a)	Formation: Barnett. Utilizing dynamic Young's modulus (E_{dyn}) and EBI from Rickman et al. (2008). Both then normalized.	EBI 7

Stage I: Data Preparation

We selected well logs that are both available across all five wells and span the targeted stratigraphic intervals. The availability across all wells was the primary criterion for inclusion. Eleven logs were chosen, including three resistivity logs with different depths of investigation. Following Hall (2016), a base-10 logarithm (log10) was calculated for each resistivity log, resulting in three additional derived logs. In total, 14 logs were used in this study. The reference estimation pipeline is described in Davy et al. (2024a), which shows the significance of the stratigraphic and petrophysical controls on the TOC and hardness reference estimation. The workflow includes the petrophysical parameters delta (δ) and cap delta (Δ), the latter of which is based on Hall (2016). The MBI, calculated from the mineral weight fractions (Fig. 3), is available for the ArcSnake and McAdam wells and will be used to estimate MBI values for the other wells.

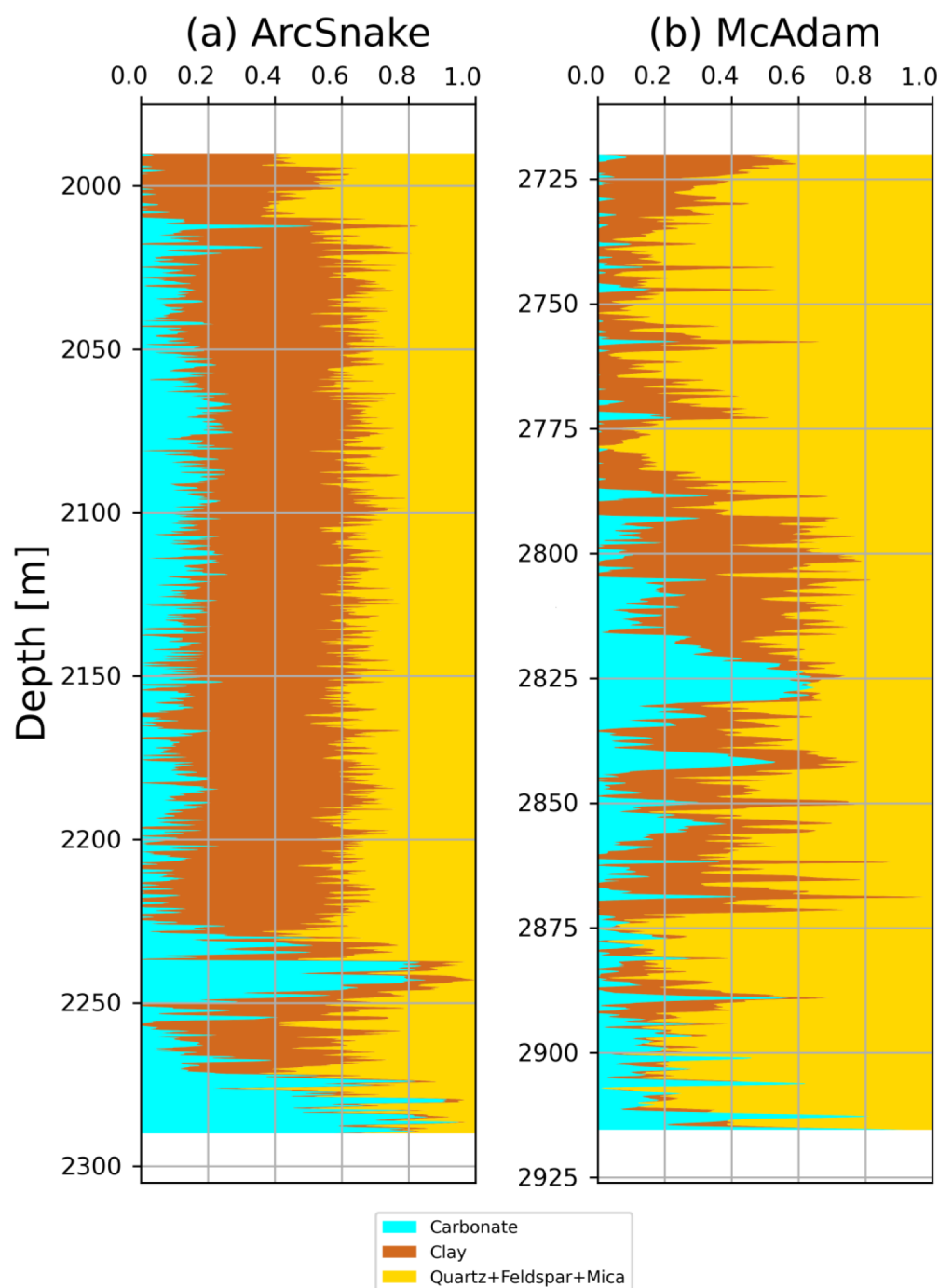


Figure 3—Stackplot of the three mineralogical data from wells (a) ArcSnake and (b) McAdam, totaling 3249 points. The data include QFM (quartz + feldspar + mica), clay, and carbonate, represented in butter yellow, chocolate brown, and turquoise, respectively. The logs are normalized, meaning the three logs are divided into fractions that sum to 1 at each point.

Stage II: Feature Selection and BI reference estimation

Feature selection based on relevance to the target variables (i.e., TOC, hardness, MBI), is an important step before applying ML models to improve prediction performance and helps understand the data better. Following standard practice across various fields, we use Pearson's correlation coefficient (R value). Based on the method from [Rong et al. \(2021\)](#) for AI-based TOC prediction, we select features with $R \geq 0.2$ or $R \leq -0.2$. Only parameters that follow this criterion are selected as the features for each prediction.

The first part of Stage II comprises the creation of TOC and hardness reference values, based on the [Davy et al. \(2024a\)](#) approach with some modifications. The modifications pertain to the selection of features

based on the R -value criterion, as illustrated in Fig. 4 for TOC (left column) and hardness (middle column). All well logs and petrophysical parameters are used for TOC prediction except the resistivity (R20, R30, R90) and photoelectric factor (PE) logs. Hardness prediction uses all 16 logs and parameters. Full details of the process are described in the referenced study and are not the primary focus of this work. The generated TOC and hardness references for all five wells are shown in Fig. 2.

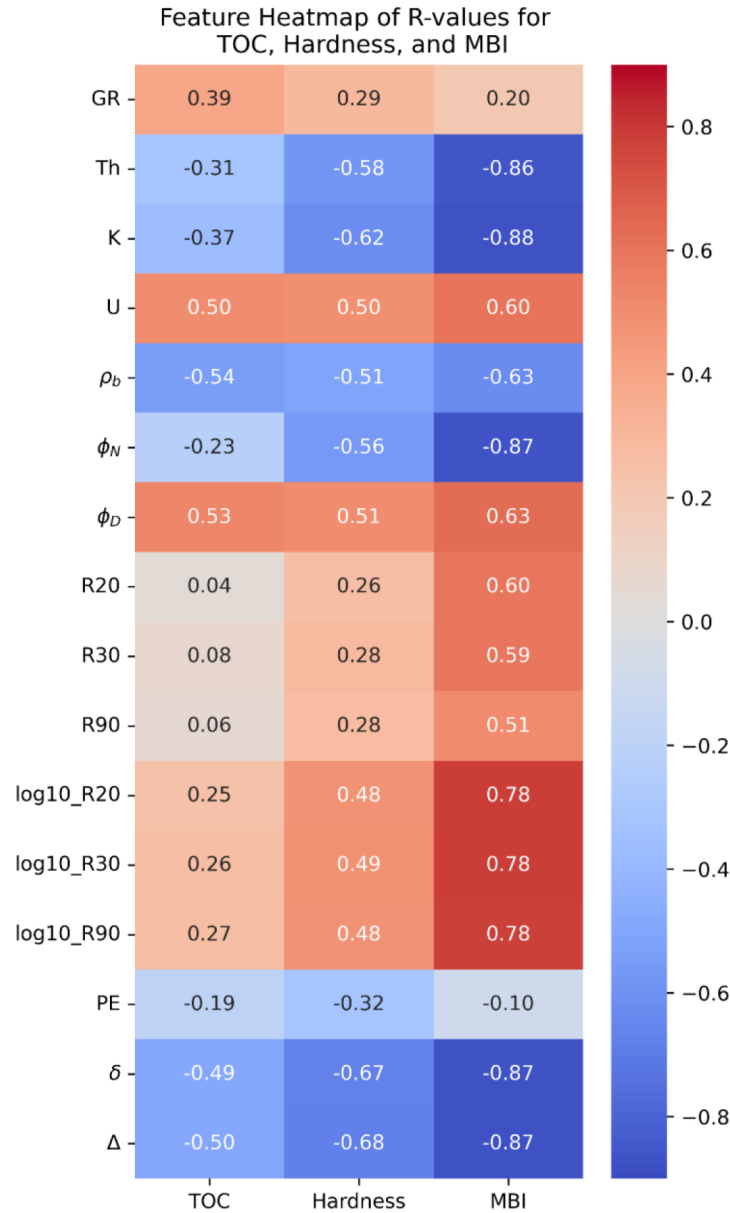


Figure 4—Correlation coefficients (R values) between the labels (TOC, Hardness, and MBI) and the 14 well logs plus 2 petrophysical features. MBI (mineralogical brittleness index) refers to the modified Jin et al. (2014a) MBI by Mews et al. (2019), denoted as "MBI 3" in this Table 2. TOC and Hardness data are available across the five wells, while the measured MBI data (not the reference MBI) is only from the two wells with mineralogical logs.

The second part of Stage II focuses on brittleness indices (BIs). We first computed a series of BIs, including the MBI, LBI, and EBI, based on equations listed in Table 2. We then calculate the R -value between each index and measured hardness, as shown in Fig. 5. To evaluate the relevance of each BI, we calculated the R -value between all 18 BIs and measured hardness. Previous studies have shown a positive linear relationship between brittleness and hardness (Lawn & Marshall, 1979), and established hardness as a reliable proxy for brittleness (Dong et al., 2017; Kim et al., 2022). Therefore, the BI that shows the highest

correlation with hardness is considered the most representative and reliable among those evaluated. This selected BI is referred to as the "selected BI" hereinafter.

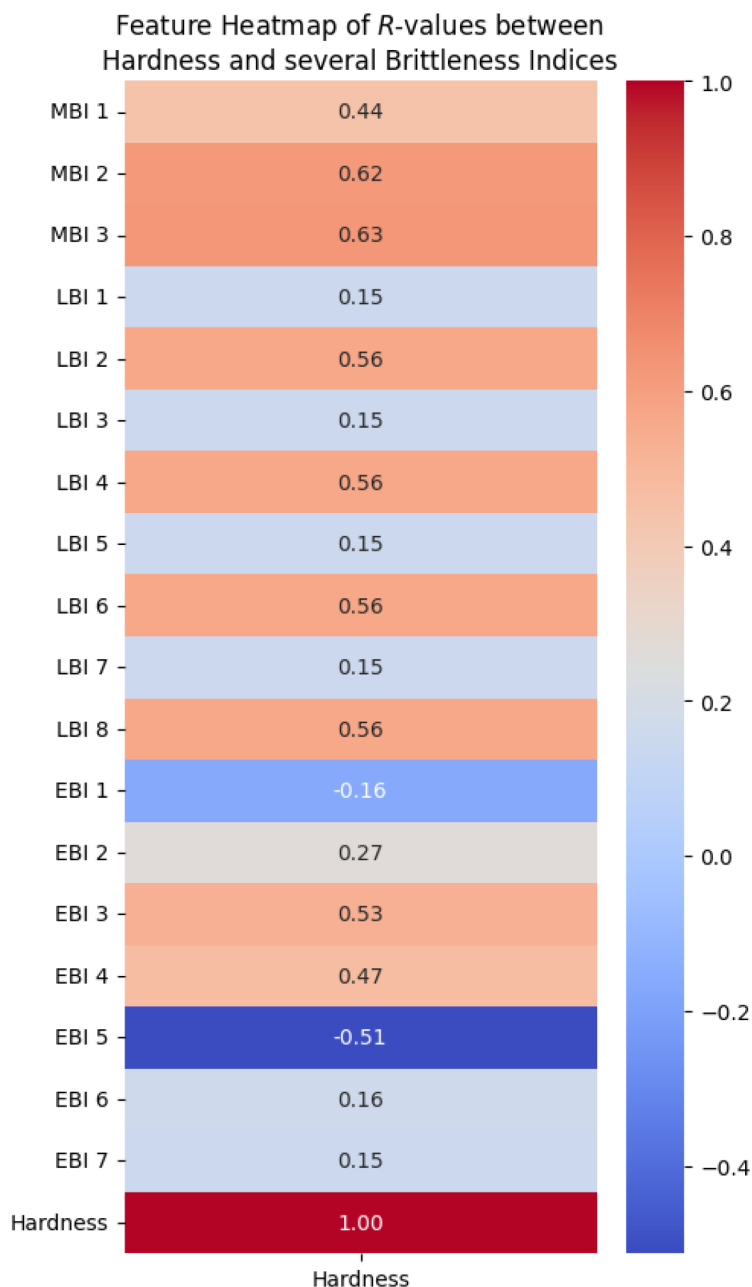


Figure 5—Correlation coefficients (R -values) between various brittleness indices (BIs, see Table 2) and hardness. A higher R -value indicates a stronger correlation, suggesting greater reliability. The most reliable mineralogical BI is MBI 3 (modified from Jin et al., 2014a) with an R -value of 0.63. Among log-based BIs, LBIs 2, 4, 6, and 8 (based on neutron porosity from Jin et al., 2014b) show the highest reliability with $R = 0.56$. The most reliable elastic BI is EBI 3 (Sun et al., 2013) with $R = 0.53$. Overall, MBI is the most reliable, while EBI is the least. Several EBIs yielded negative R -values, indicating unrealistic inverse correlations with hardness and thus are considered unreliable.

Since hardness is a reliable indicator of brittleness, it can be used to estimate brittleness indices (BI). A linear trend is generally expected when cross-plotting hardness against a selected BI. However, the relationship often exhibits complexity, which can be confirmed by the limitations of traditional regression methods. Using hardness alone does not provide accurate BI estimates. To address this, we employ a data-driven AI-based curve fitting method, to incorporate additional relevant parameters. This approach follows

the framework of [Davy et al. \(2024a\)](#) that utilized petrophysical and geological constraints, and is used to generate BI references across all five wells in this study.

Stage III: Cross-plot Analysis

Cross-plot analysis serves as a valuable tool in geophysics for combining multiple parameter datasets and uncovering patterns or outliers in reservoir properties. This study focuses on three primary parameters: TOC, hardness, and the selected BI. We analyze the relationship between hardness and BI. We use TOC, relevant well logs, and petrophysical parameters as color codes to reveal trends linked to organic matter and clay content in the target interval.

Afterward, we integrate TOC with other available well logs such as Gamma Ray (GR), Uranium (U), Thorium (Th), and petrophysical parameters δ and Δ . From [Glover \(2005\)](#), Thorium is directly related to clay percentage, and Uranium to organic matter. The δ parameter, as described by [Davy et al. \(2024a\)](#), has positive values in the presence of clay and negative values in the presence of organic matter. The Δ parameter introduced in this study shows a similar pattern to δ . We also include a cross plot of hardness versus the selected BI, indicating data points with TOC values above or below 3 wt%. The 3 wt% value is chosen since it is the average TOC percentage of the basin ([Dong et al., 2018](#); [Ayranci et al. 2018a,b](#)).

We also analyze cross plots of hardness against mineralogical data, limited to the two wells with available mineralogical data. There are three mineralogical data: quartz-feldspar-mica (QFM), clay, and carbonate. This results in three cross plots—hardness versus QFM, clay, and carbonate. All are plotted using TOC as the color code to observe the relationship between mineralogy, TOC, and hardness of unconventional reservoirs.

Result

Selected BI reference estimation

As shown in [Fig. 5](#), MBI 3 (from [Jin et al., 2014a](#), modified by [Mews et al., 2019](#)) and MBI 2 (from [Jin et al., 2014a](#)) show the highest R -values with hardness, at 0.63 and 0.62, respectively. LBIs 2, 4, 6, and 8, all based on ϕ_N , as noted by [Jin et al. \(2014b\)](#), each yield an R -value of 0.56. EBI 3, from [Sun et al. \(2013\)](#), gives an R -value of 0.53. All MBIs and LBIs show positive correlations with hardness, consistent with the known positive relationship ([Lawn & Marshall, 1979](#)) and confirming hardness as a reliable proxy for brittleness ([Kim et al., 2022](#)). In contrast, some EBIs—such as EBI 1 ([Rickman et al., 2008](#)) and EBI 5 ([Jin et al., 2014a](#))—produce negative R -values, indicating an unrealistic inverse relationship with hardness and making them less suitable. However, [Dong et al. \(2017\)](#) also showed that hardness is positively correlated with Young's modulus and negatively correlated with Poisson's ratio. Based on its stronger correlation with hardness compared to LBI and EBI, MBI 3 or simply "MBI," is selected as the BI for this study.

Since hardness is a reliable indicator of brittleness in the Horn River shales ([Dong et al., 2017](#); [Kim et al., 2022](#)), it can be used to estimate a brittleness reference. A cross plot of measured hardness and MBI ([Fig. 6](#), limited to the two wells with mineralogical data, as shown in [Fig. 3](#)) confirms a linear relationship with R of 0.63. However, traditional linear, quadratic, logarithmic, and power regression models used to predict MBI from hardness ([Fig. 7](#)) produce R^2 values below 0.40. This indicates that while related, hardness alone is insufficient for accurate MBI prediction. The scattered data further suggests additional influencing factors. Moreover, mineralogical data required to calculate MBI are only available for ArcSnake and McAdam wells. Given the weak performance of traditional regression, the lack of empirical equations linking hardness and MBI, and limited MBI data coverage, we apply machine learning to estimate the MBI reference. This approach is based on the AI-driven method proposed by [Davy et al. \(2024a, b\)](#), using hardness and other relevant well logs and parameters.

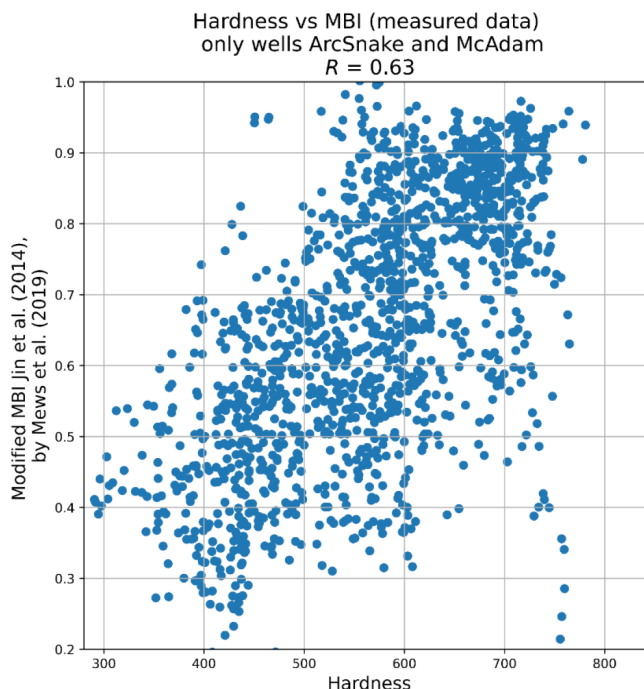


Figure 6—Crossplot of measured hardness versus MBI (Jin et al., 2014; modified by Mews et al., 2019) for the two wells with measured mineralogical logs. The linear trend indicates a positive relationship between hardness and MBI, evidenced by a $R = 0.63$. However, the scatter of data points highlights complexities beyond hardness, suggesting additional factors influencing MBI.

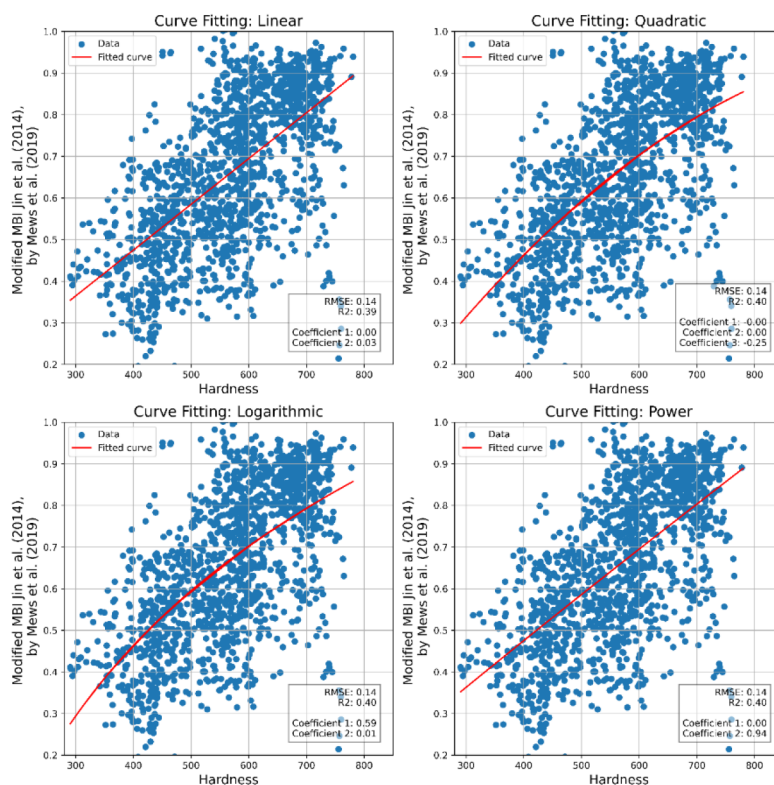


Figure 7—Comparison of traditional regression models (linear, quadratic, logarithmic, and power) applied to the hardness versus MBI crossplot (refer to Fig. 6). All models yielded unsatisfactory results ($R^2 < 0.40$) due to the scattered data points, indicating that hardness alone is insufficient to predict MBI accurately. This suggests the need for additional input variables and the potential superiority of advanced approaches, such as artificial intelligence (AI) / Machine Learning (ML), over traditional regression methods.

To estimate the MBI reference, we use the same feature selection process described earlier, based on R -values. As shown on the right side of Fig. 4, all parameters meet the selection threshold except for the PE log, which has an R -value of -0.10 and is excluded. This leaves 15 relevant inputs: 13 well logs and 2 petrophysical parameters. Hardness, with an R -value of 0.63 with MBI, is also included, bringing the total to 16 inputs. Using the same framework applied for TOC and hardness estimation, MBI trends across the five wells are generated and shown in Fig. 8.

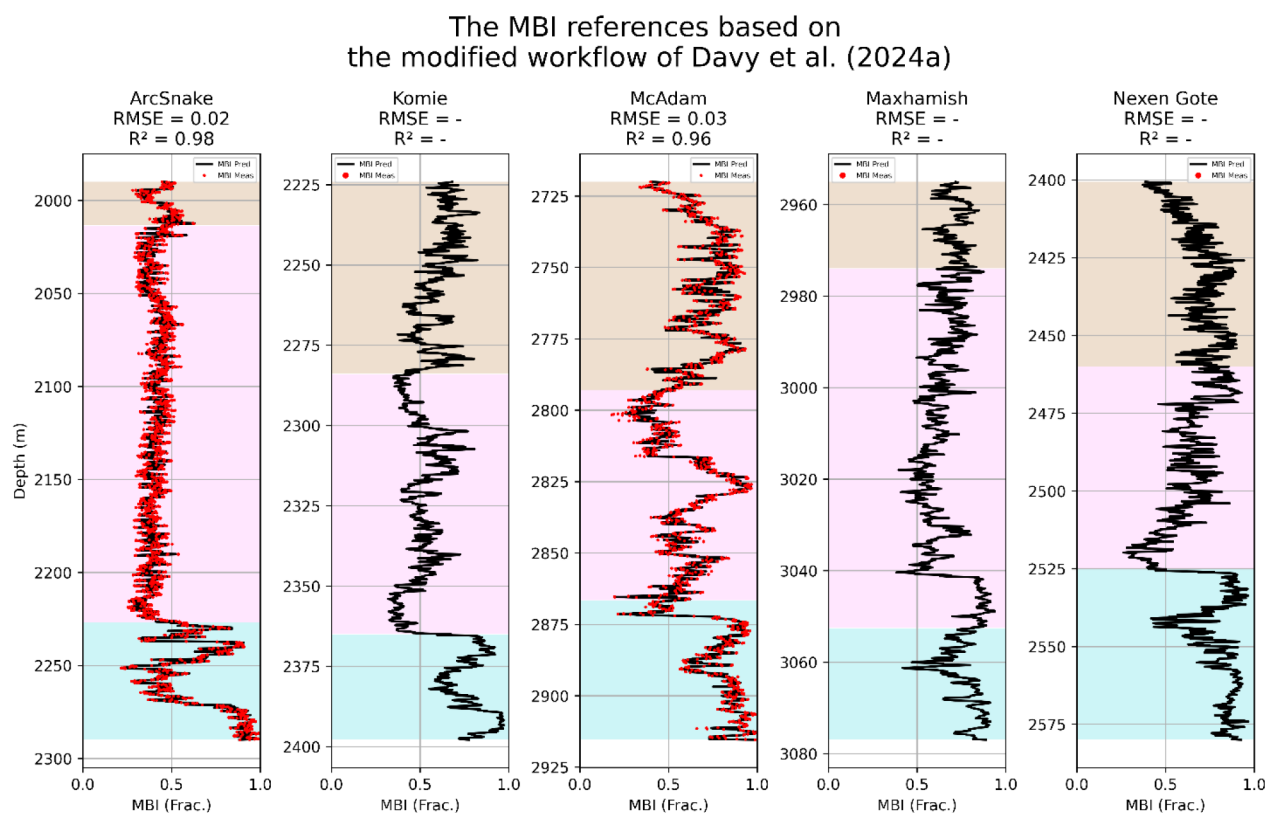


Figure 8—The references for the MBI (MBI 3 as in the Table 2) predictions based on the workflow of Davy et al. (2024a) for each of the wells. The RMSE (root-mean-square error) and R^2 (coefficient of determination) values are included. The wells with available mineralogical data (ArcSnake and McAdam) are used for training, and the optimized model derived from this training is subsequently applied to estimate the MBI for the remaining wells. Red dots represent measured (actual) values, and black curves represent predicted values. Background colors indicate stratigraphic units of prospective intervals in the basin: brown for the Muskwa formation, magenta for the Otter Park member, and cyan for the Evie member, from the most recent to the oldest.

Hardness-MBI and Hardness-Mineralogical Cross-plot Analyses

After generating the reference values for TOC, hardness, and MBI across the five wells (Figs. 2a, 2b, and 8, respectively), we performed two cross-plot analyses. The first compares reference hardness with reference MBI, while the second compares measured hardness with mineralogical data from the two wells that contain such data.

The first cross plot (Fig. 9a) shows a similar trend to the measured hardness versus measured MBI cross plot for the two wells with mineralogical data (Fig. 6). Both plots suggest a linear relationship, supported by R -values of 0.63 between the measured data and 0.77 between the reference data. We also introduce a third variable into the cross plots by using color coding to represent various well logs and petrophysical parameters, as shown in Figs. 9 and 10.

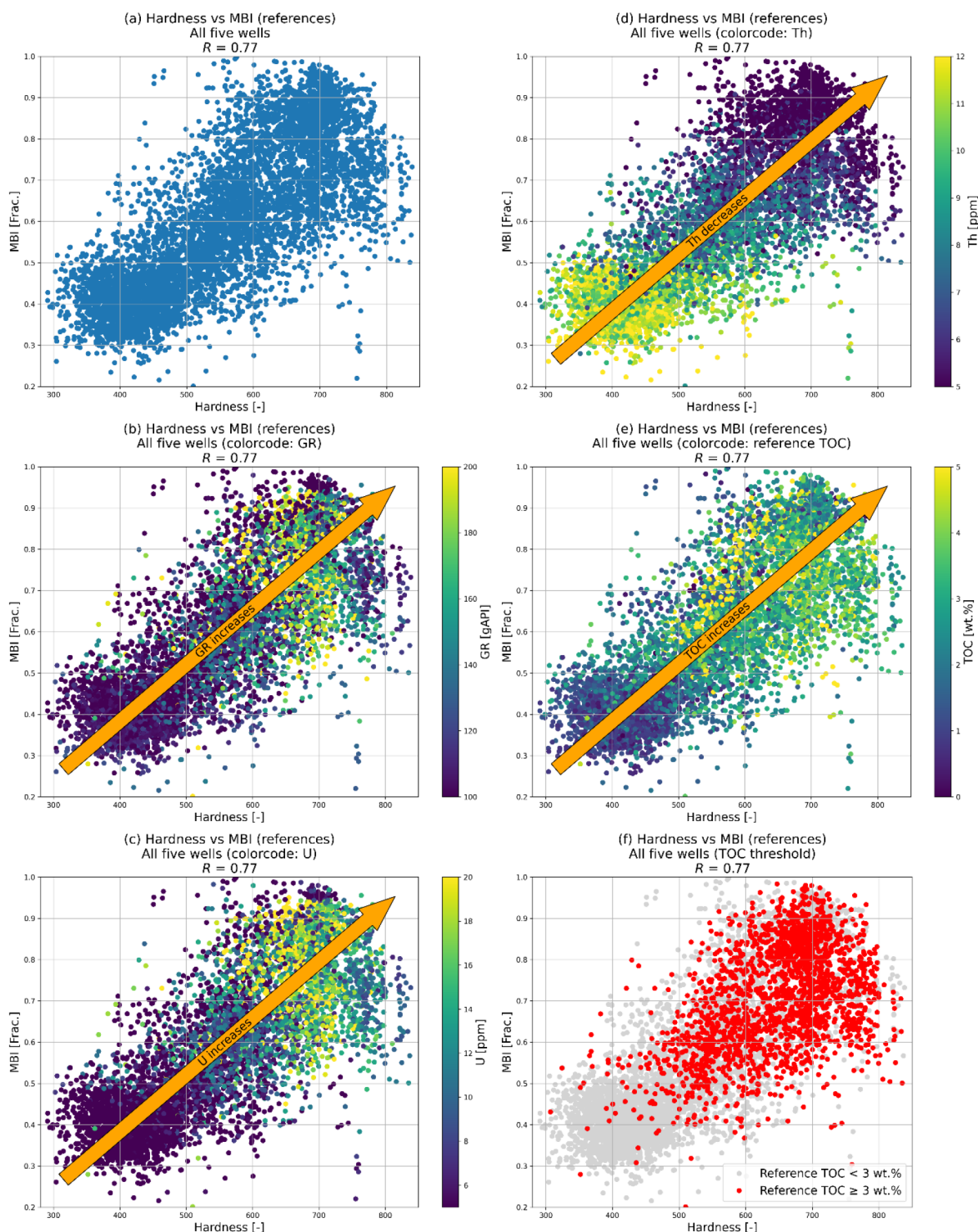


Figure 9—(a) Cross plot of reference hardness versus reference MBI for the five wells; the R -value here is 0.77, compared to an R -value of 0.63 for measured hardness and measured MBI on the two wells that have mineralogical data (Figs. 6 and 7). The color coding in the plots is for the following well logs/parameters: (b) Gamma Ray, (c) Uranium count, (D) Thorium count, and [e] reference TOC. Points are distinguished between prospective and non-prospective points with respect to a TOC threshold of 3 wt%, as shown in (f). In all subfigures [b] through [e], the orange arrow shows trends with increasing hardness and MBI, reflecting corresponding changes in the color-coded parameter.

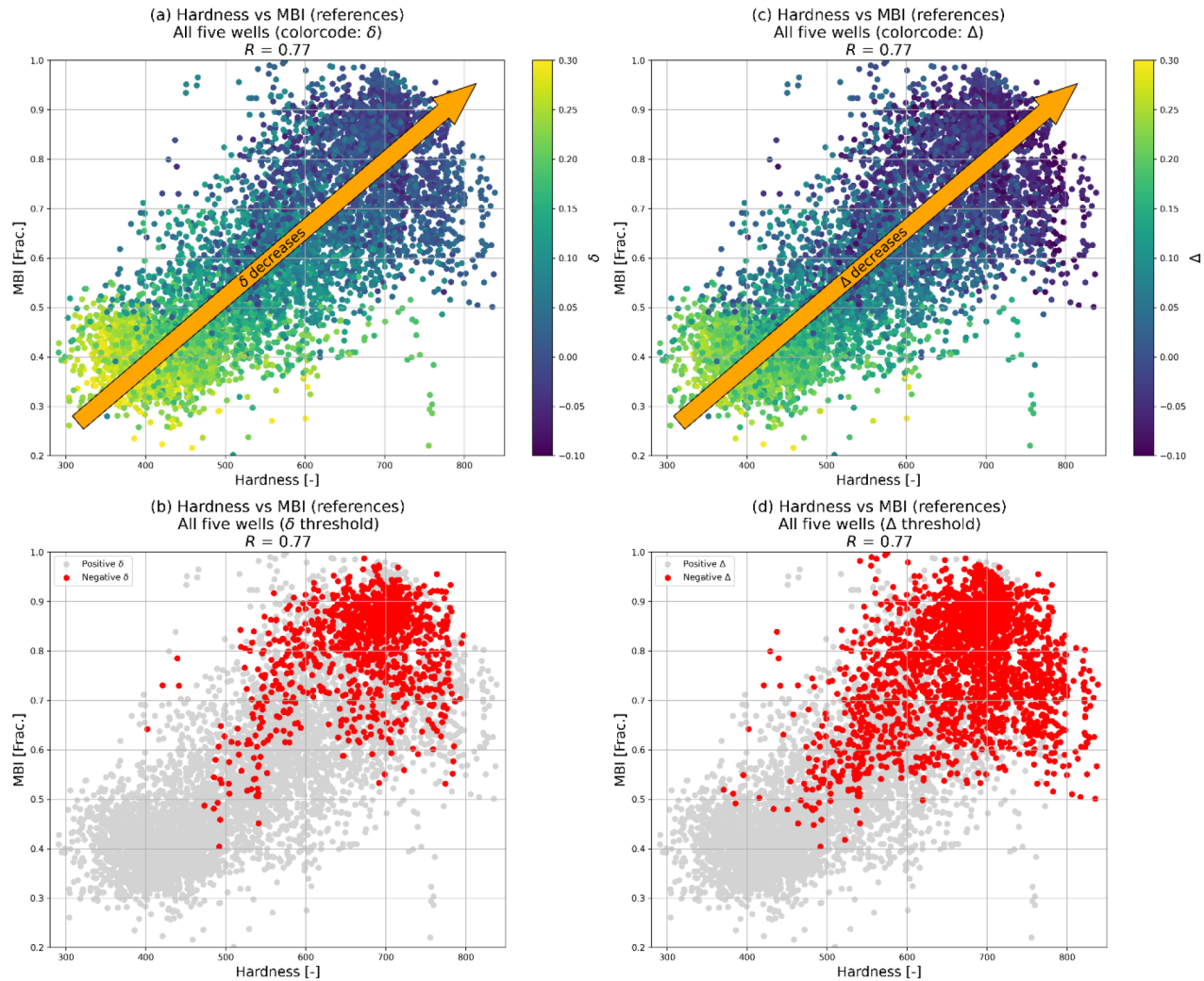


Figure 10—Cross plot of reference hardness versus reference MBI for the five wells. Color indicates petrophysical parameters. Subfigures (a) and (b) use the difference between neutron porosity and bulk density (δ), while subfigures (c) and (d) use the difference between neutron porosity and density porosity (Δ). Subfigures (a) and (c) use their respective petrophysical parameters as the color code, while subfigures (b) and (d) threshold these parameters to positive or negative values. Arrows in subfigures (a) and (c) show the trend of changes in the respective petrophysical parameters with increasing hardness and MBI.

When Gamma Ray (GR) is used as the color code (Fig. 9b), it increases with both hardness and MBI. Similar patterns are seen for Uranium (U) and TOC (Figs. 9c and 9e), showing positive relationships with hardness and MBI. In contrast, Thorium (Th), shown in Fig. 9d, shows a negative correlation: hardness and MBI increase as Th decreases. Using an average TOC value of about 3 wt.% for the basin, we classify points as prospective (TOC $\geq$ 3 wt.%, red dots) or non-prospective (TOC < 3 wt.%, transparent gray dots), as shown in Fig. 9f. Most prospective points fall in the upper right quadrant, where hardness and MBI are highest. The petrophysical parameters δ and Δ (Figs. 10a and 10c) show a decreasing trend with increasing hardness and MBI. Their negative values are concentrated in the upper right quadrant of the cross plot, as seen in Figs. 10b and 10d.

The second cross-plot analysis (Fig. 11) examines individual mineralogical data for ArcSnake and McAdam—normalized fractions of QFM (quartz, feldspar, and mica), clay, and carbonate (see Fig. 3)—against reference hardness. QFM shows a positive correlation with hardness (Figs. 11a and 11d), while clay shows a negative correlation (Figs. 11b and 11e). Carbonate, however, shows no clear trend with hardness (Figs. 11c and 11f), suggesting little or no relationship. TOC increases with QFM content, causing prospective points to cluster in the upper right of the hardness–QFM plot (Fig. 11d). In contrast, TOC

decreases with clay, leading to prospective points clustering in the lower right of the hardness–clay plot (Fig. 11e).

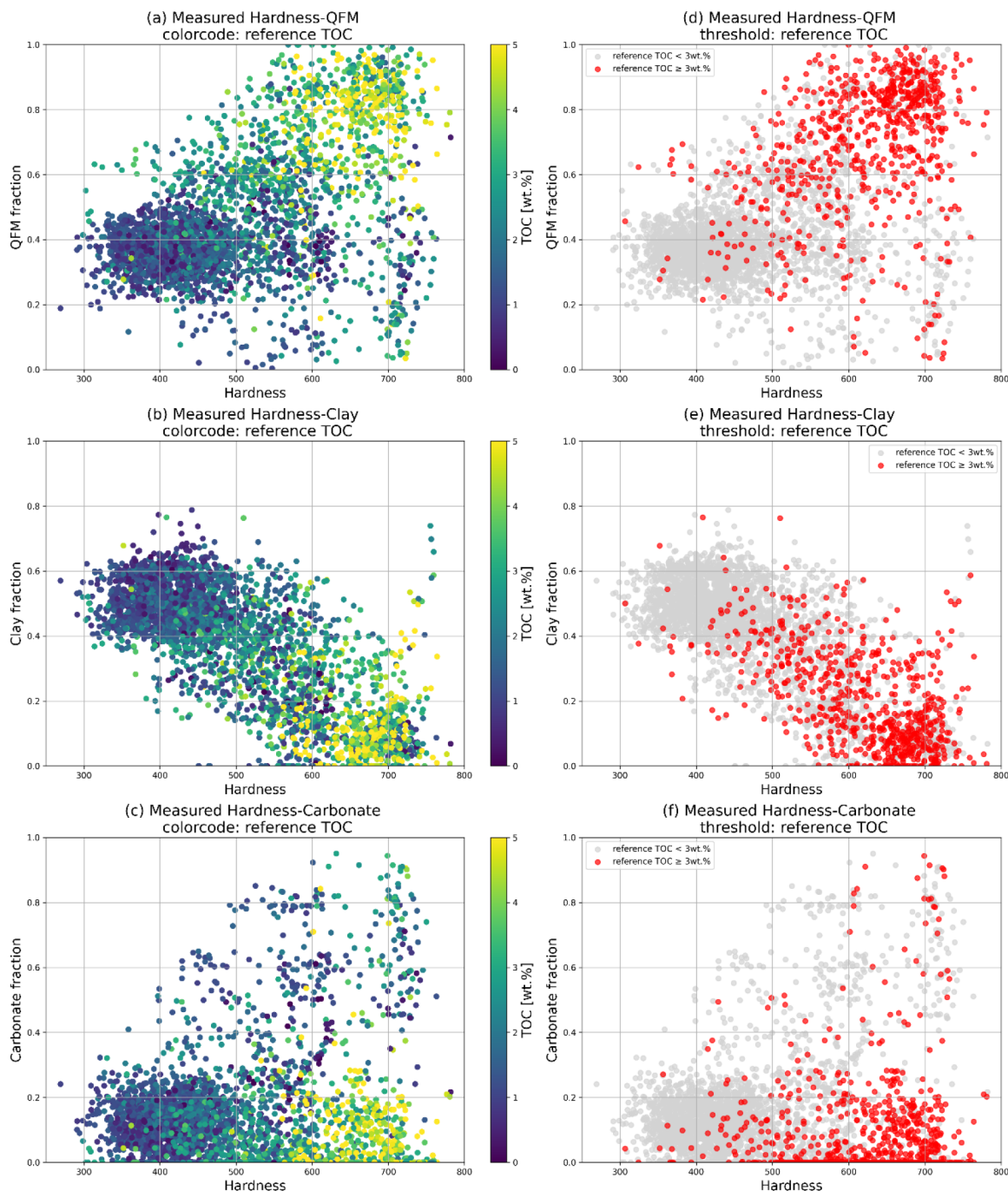


Figure 11—Cross-plots of predicted hardness versus individual normalized mineralogical logs (Fig. 3). Only data points with both predicted hardness and corresponding mineralogical logs are plotted. Subfigures (a) and (d) show the hardness versus QFM (quartz, feldspar, and mica). Subfigures (b) and (e) show the hardness versus clay. Subfigures (c) and (f) show the hardness versus carbonate. Cross-plots on the left side use predicted TOC as the color code, while those on the right side use the reference TOC threshold of 3 wt.%. Clear relationships are observed between hardness and QFM, as well as hardness and clay, while no clear relationship is observed between hardness and carbonate.

Discussion

The comparison of various brittleness indices (BIs), as summarized by Mews et al. (2019), includes the mineralogical-based brittleness index (MBI), log-based brittleness index (LBI), and elastic brittleness index

(EBI). This comparison evaluates the correlation (R -values) between each BI and hardness, as shown in Fig. 5. The top-performing indices for each type are: (1) MBI by Jin et al. (2014a), modified by Mews et al. (2019) (MBI 3 in Table 2), with the highest R -value of 0.63; (2) LBI by Jin et al. (2014b), developed empirically for several formations using ϕ_N (LBIs 2, 4, 6, and 8 in Table 2); and (3) EBI by Sun et al. (2013), labeled EBI 3 in Table 2. Since brittleness and hardness are known to be linearly related (Altindag & Guney, 2010) and hardness is a reliable proxy for brittleness in the Horn River shale (Dong et al., 2017; Kim et al., 2022), negative R -values are not expected. All MBIs and LBIs show positive R -values with hardness. However, some EBIs, such as EBI 1 (Rickman et al., 2008) and EBI 5 (Jin et al., 2014a), yield negative R -values, which are inconsistent with expected relationships and suggest that EBIs are generally less reliable.

While comparing MBI with LBI, the MBI is often considered a more direct indicator of brittleness due to its reliance on the mineralogical distinction between typically brittle minerals (e.g., quartz) and ductile minerals (e.g., clay). That said, this relationship is not universally straightforward; for example, Dong et al. (2017) demonstrated that only quartz cement shows a strong correlation with brittleness, whereas detrital quartz does not. LBI, however, is entirely empirical and depends on ϕ_N (neutron porosity) that is formulated for specific formations. Porosity alone is not a reliable indicator of brittleness because the two are not directly related. For example, a quartz-rich interval (e.g., sandstone) may have high or low porosity due to factors such as depth, pressure, or compaction—none of which directly reflect the brittleness of the rock. However, a high concentration of brittle minerals like quartz generally indicates higher brittleness, making mineralogy a more robust basis for brittleness estimation.

A key issue with EBI becomes clear when comparing the P-wave velocity (V_p) of quartz and carbonate (e.g., calcite). According to Mavko et al. (2009), quartz has a V_p of up to 6.06 km/s, while calcite reaches up to 6.64 km/s. Both are higher than clay minerals, which vary from roughly 1.44 km/s for kaolinite to 4.32 km/s for illite. While high V_p in quartz and low V_p in clays may correspond to high and low brittleness, respectively, the high V_p of carbonate does not necessarily imply high brittleness. In fact, studies by Alzahabi et al. (2015) and Rybacki et al. (2016) show that higher carbonate content often results in lower brittleness. This mismatch between high elastic velocity and uncertain brittleness in carbonates makes EBI less reliable.

It is also important to note that although MBI is generally considered more reliable than LBI or EBI, MBI 1 (Jarvie et al., 2007) shows a lower R -value compared to several LBIs. This outcome is likely due to its sensitivity to carbonate content. MBI 1 is known to perform well in formations with low carbonate content. However, our dataset includes a significant amount of carbonate, which causes the brittleness estimated by MBI 1 to be underestimated and results in a weaker correlation with hardness. Furthermore, MBI 1 includes carbonate in the denominator, implying that carbonate is considered a ductile component. In contrast, the behavior of carbonate in our data is more consistent with brittle minerals. This characteristic is better captured by the formulations of MBI 2 and MBI 3, which consequently show higher R -values with hardness.

After identifying the modified MBI approach by Jin et al. (2014a), or MBI 3, as the most reliable method for estimating brittleness, we noted that mineralogical data (Fig. 3) are available for only two of the five wells. Fig. 7 shows our attempt to estimate the MBI reference from hardness using multiple regression models, based on the expected positive correlation between brittleness and hardness. However, this approach was ineffective due to significant scatter in the MBI–hardness cross-plot, suggesting that the relationship is more complex and cannot be explained by hardness alone. Since no direct empirical equations exist to link MBI and hardness, and additional input is clearly needed, we adopted an AI-based method following Davy et al. (2024a) to improve the estimation. To choose relevant features, we used a correlation threshold (R -value ≥ 0.2 or ≤ -0.2), as adopted by Rong et al. (2021). This produced 16 features: hardness, thirteen well logs, and two petrophysical parameters (see right column of Fig. 4). Using this input and the two wells with mineralogical data, we generated the final MBI reference values for all five wells.

We observed clear trends in the cross-plot of reference hardness versus reference MBI using different well logs and parameters as color codes. When gamma ray (GR) is used (Fig. 9b), it increases with both MBI and hardness, suggesting a linear relationship. However, rising GR values may indicate either more clay (Serra et al., 1980) or more organic matter (Passey et al., 1990). To distinguish between the two, we applied uranium (U) and thorium (Th) as color codes. U correlates with organic matter, while Th correlates with clay content (Glover, 2005). When U is used (Fig. 9c), it increases with hardness and MBI. In contrast, Th decreases with both (Fig. 9d). The U trend suggests organic-rich intervals in the high hardness–MBI region, which is supported by the TOC trend (Fig. 9e), showing similar behavior. Low hardness and MBI values are associated with higher Th, suggesting increased clay content in these zones.

In Fig. 9f, we separate prospective ($\text{TOC} \geq 3 \text{ wt.}\%$) and non-prospective ($\text{TOC} < 3 \text{ wt.}\%$) points. The prospective points cluster in the upper right of the plot, indicating a positive relationship between TOC and brittleness in our dataset. This relationship is likely indirect and may reflect mineralogical or diagenetic factors. For example, Mo et al. (2018) noted that high TOC intervals often coincide with higher silica content, which can enhance brittleness. Although Moghadam et al. (2019) reported a more complex relationship, primarily observing an inverse correlation between clay content and brittleness, our results suggest that high TOC may be associated with conditions that promote greater brittleness. The petrophysical properties δ and Δ (as plotted in Figs. 10a and 10c) exhibit a trend of decreasing with increasing hardness and MBI. This aligns with the δ behavior based on the neutron porosity (ϕ_N) – bulk density (ρ_b) crossover, as explained in Davy et al. (2024a). The δ is negative in organic-rich zones and positive in clay-rich zones; Δ shows a similar trend. Plotted based on sign (Figs. 10b and 10d), negative values are clustered in the top right of the cross-plot, which helps to identify prospective intervals.

Fig. 11 shows noticeable trends in the cross-plots of hardness versus individual mineralogical fractions. The positive correlations between QFM (quartz, feldspar, mica), hardness, and TOC (Figs. 11a and 11d) are consistent with the idea that specific silica-bearing components, particularly quartz cement, contribute to increased brittleness and hardness. While the role of detrital quartz, feldspar, and mica is more complex, the observed trends suggest a mineralogical influence on mechanical properties. In the Horn River Basin, this also corresponds to higher TOC. In contrast, higher clay content is associated with lower brittleness and hardness, as shown in Figs. 11b and 11e, due to the ductile nature of clay. A decrease in clay content often indicates more organic matter, explaining the negative relationship between clay fraction and TOC. The carbonate fraction (Figs. 11c and 11f) does not show any specific trend with either hardness or TOC, i.e., it has little influence on brittleness or organic content. According to Dong et al. (2017), quartz in Horn River shale generally develops from the recrystallization of biogenic silica, while feldspars and micas remain detrital, like clay. If hardness values had been plotted against quartz alone, there would probably have been a stronger correlation. However, no such data exist for quartz alone, only for the combined QFM. They also commented that the Horn River Shale carbonate minerals are not load-bearing components.

Overall, the hardness–MBI cross-plot is helpful in characterizing unconventional resources and revealing patterns related to organic matter and clay content. These trends may also reflect depositional environment, as the five wells span a proximal-to-distal gradient, influencing mineral distribution and organic matter enrichment. The cross-plots of hardness versus individual mineralogical fractions highlight quartz and clay fractions as important indicators for unconventional reservoir potential, while carbonate appears to play a minimal role.

Conclusion

The study illustrates that the mineralogical-based brittleness index (MBI) is the preferable choice over the log-based brittleness index (LBI) and the elastic brittleness index (EBI). The MBI is more reliable, as it indicates a higher correlation with hardness and can distinguish between brittle and ductile minerals. The LBI tends to be more empirical, whereas the EBI is inconsistent (i.e., shows a negative R -value to hardness).

The cross-plot of MBI versus hardness exhibits a generally linear trend, although with significant scatter. This scatter likely reflects a combination of mineralogical, textural, and organic richness complexities. For example, as shown by Dong et al. (2017), only authigenic (cement) quartz is strongly associated with increased brittleness, whereas detrital quartz has little or no effect. Importantly, while MBI is volumetric, hardness is influenced by geometry—a factor highlighted in several studies (e.g., El-Husseiny et al., 2019; Jaballah et al., 2021; Reijmer et al., 2021; Salih et al., 2023). Thus, although MBI is strongly correlated with hardness, it does not incorporate geometric effects (i.e., the fabric or cementation of the rock itself), which may be partially captured by EBI. In recognizing that hardness and elastic stiffness are not identical, future brittleness evaluations need to more specifically incorporate geometric effects.

Color-coded cross-plots (e.g., using GR, U, Th, TOC, δ , or Δ) improve the interpretability of MBI–hardness trends. The prospective zones (TOC $\geq$ 3 wt.%, low clay) cluster in the upper right with increasing U and negative δ or Δ . The non-prospective, clay-rich zones (TOC < 3 wt.%) appear in the lower left with higher Th and positive δ or Δ . Quartz and clay contents are primary indicators for brittleness, whereas carbonates play a minor role. These observations are consistent with Dong et al. (2017).

Future work might integrate porosity and resistivity into cross-plot analyses and explore applying the MBI–hardness framework in diverse geological settings. Where direct hardness data are unavailable, inverted elastic parameters may serve as proxies for mapping brittleness and identifying favorable unconventional reservoir intervals (i.e. sweet spot).

Data Availability

The dataset used in this study, obtained from the five wells in the Horn River Basin, is classified as confidential and cannot be shared. Furthermore, the Python notebooks employed for analysis and visualization are not disclosed and are considered the author's intellectual property. To make enquiries or engage in discussions, please contact the corresponding author by email at nanditodavy96@gmail.com or g202115270@kfupm.edu.sa.

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